

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester of B. Tech (EC) Examination****May 2015****EC312 Digital Signal Processing****Date: 14.05.2015, Thursday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below.**

- a. What is basically Digital Signal Processing? [01]
- b. Describe advantages of Digital Signal Processing over Analog Signal Processing. [02]
- c. Determine if the systems described by the following input-output equations are Time Variant or Time Invariant. (i) $y(n) = \cos[x(n)]$ (ii) $y(n) = x(2n)$ [02]
- d. Determine if the systems described by the following input-output equations are Linear or Non-linear. (i) $y(n) = x(-n)$ (ii) $y(n) = x(n^2)$ [02]

Q – 2.a Define and Prove Stability and Causality of Discrete Time Systems. [04]**Q – 2.b Answer any two questions. [10]**

- (i) Explain Sampling Process with necessary figures.
- (ii) Compute the convolution $y(n)$ sequence for the following pair of signals.

$$x(n) = \{ \underset{\uparrow}{1}, 2, 4 \} \text{ and } h(n) = \{ \underset{\uparrow}{1}, 1, 1, 1, 1 \}$$
- (iii) Prove that $TD [FD(x[n])] \neq FD [TD(x[n])]$, where TD is Time Delay, FD is Folding and $x[n]$ is discrete time signal.

Q - 3 Answer any two questions. [14]

- a. Specify the properties of Region of Convergence (ROC). Prove any two properties of Z-Transform with necessary examples.
- b. Determine Z-transform and sketch the ROC of following sequence.

$$x[n] = \left(\frac{-1}{9}\right)^n u(n) - \left(\frac{1}{4}\right)^n u(-n-1)$$

- c. Find Inverse Z-Transform of given $X[z]$ using Partial Fraction Expansion Method.

$$X[z] = \frac{z^2}{[z - 0.5][z - 1]^2}$$

SECTION – II**Q - 4 Answer the question below.**

- a. Define Digital Filter. [01]
- b. Write down the advantages of Digital Filter over Analog Filter. [02]
- c. Enlist Applications of Digital Signal Processing [02]
- d. Write Formula for Discrete Fourier Transform (DFT) and Inverse Discrete Fourier Transform (IDFT). [02]

Q – 5.a Explain Bilinear Z Transform Method of Infinite Impulse Response Filter Design. [07]**OR****Q – 5.a** Describe Different Types of Windows functions with mathematical equation and spectrum for designing Finite Impulse Response Filter. [07]**Q – 5.b** Obtain Direct Form-I, Direct Form-II, Cascade Form and Parallel Form Realizations [07]

for this system
$$y(n) = \frac{3}{4}y(n-1) - \frac{1}{8}y(n-2) + x(n) + \frac{1}{3}x(n-1)$$

Q – 6.a Perform Circular Convolution of given two sequences. [07]

$$x[n] = \{ \underset{\uparrow}{2}, 1, 2, 1 \} \text{ and } h[n] = \{ \underset{\uparrow}{1}, 2, 3, 4 \}.$$

Q – 6.b If $X[n]$ is given as $\{1, 2, 2, 1\}$. Find corresponding Discrete Fourier Transform (DFT) of $X[k]$ using Decimation In Time Fast Fourier Transform (DIT FFT) Algorithm. [07]**OR****Q – 6.a** If $X[n]$ is given as $\{1, 2, 2, 1, 0, 0, 0, 0\}$. By using butterfly structure of Decimation in Frequency (DIF), Calculate output for the each stage. [07]

Q – 6.c Compute 4 point DFT of following sequences. Use matrix for DFT computation [07]

- (i) $x[n] = \{2, 0, 0, 1\}$
- (ii) $h[n] = \{4, 3, 2, 1\}$
